# Pedro Brunialti Lima de Andrade

Office; 119 Kerchof Hall, 141 Cabell Drive, Charlottesville, VA 22904, USA
eyu4dh@virginia.edu pbrunialti.github.io/

## Areas of interest

Homotopy Theory, Algebraic Topology, and Mathematical Physics

## Education

**University of Virginia** **Charlottesville**
*Title: Involutions on $(\infty, n)$-categories of bordisms and reflection TQFTs* *2022-Present*
Ph.D. in Mathematics – Advisor: Julie Bergner, expected May 2027

**Federal University of Minas Gerais - UFMG** **Belo Horizonte**
*Title: Comparing Models for $(\infty, 1)$-categories.* *2020-2022*
M.Sc. in Mathematics – Advisor: Rodney Josué Biezuner

**Federal University of Minas Gerais - UFMG** **Belo Horizonte**
*B.Sc. in Physics* *2016-2019*

## Papers and preprints

**Articles**

[1] Abdelmalek Abdesselam, Pedro Brunialti, Tristan Doan, and Philip Velie. A bijection for tuples of commuting permutations and a log-concavity conjecture. *Res. Number Theory*, 10(2):Paper No. 45, 10, 2024.

[2] Julie Bergner, Pedro Brunialti, Eleftherios Chatzitheodoridis, and Josh Turner. Trees associated with unitary partition complexes, in preparation.

[3] Anna Marie Bohmann, Pedro Brunialti, Bertrand Guillou, David Mehrle, and Chris Portwood. Free bi-incomplete Tambara functors are almost never free, in preparation.

[4] Pedro Brunialti. Involutions on $(\infty, n)$-categories of bordisms and reflection TQFTs, in preparation.

## Invited talks

**Canadian Mathematical Society Winter Meeting**
*Involutions on $(\infty, n)$-categories of bordisms and reflection TQFTs* *(planned) 12/2026*
Montréal, Québec

**Geometry and Topology Seminar**
*Involutions on $(\infty, n)$-categories of bordisms and reflection TQFTs* *09/2026*
Virginia Commonwealth University, Richmond

**5 College Geometry-Topology Seminar**
*Free bi-incomplete Tambara functors are almost never flat* *04/2026*
University of Massachusetts, Amherst

## Contributed talks

**2026 Fall Eastern Sectional Meeting**
*Involutions on $(\infty, n)$-categories of bordisms and reflection TQFTs* *10/2026*

**Union College Mathematics Conference**
*Involutions on $(\infty, n)$-categories of bordisms and reflection TQFTs* *09/2026*

**Talbot Workshop**
*Walker–Wang models and braided categorical nets* *06/2026*

**Higher Differential Geometry**
*$(\infty, n)$-symmetric monoidal categories with involutions* *05/2026*
University of Greifswald, Germany

**UVA Topology Seminar**
*$(\infty, n)$-symmetric monoidal categories with involutions* *02/2026*

**BUGCAT Conference**
*$(\infty, 2)$-symmetric monoidal categories with involutions* *11/2025*
Binghamton University, USA

**UVA Graduate Seminar**
*Chern classes in physics* *09/2025*

**UVA Graduate Seminar**
*Understanding and classifying TQFTs* *08/2024*

**UVA Graduate Seminar**
*What are universal properties?* *01/2024*

**UVA Graduate Seminar**
*A mathematical formulation of quantum mechanics* *01/2023*

**I Mathematics Meeting of Minas Gerais**
*Equivalence between homotopy theories* *09/2021*

**First Brazilian Meeting on Category Theory**
*Modern homotopy theory: model categories* *01/2021*

**First Brazilian Meeting on Category Theory**
*Tutorial: categories and functors* *01/2021*

**Group for Fundamental Theory (UFMG)**
*Interpreting the measurement axiom* *06/2021*

**Group for Fundamental Theory (UFMG)**
*The mathematics of gauge theories* *06/2021*

**Math-Phys-Cat group**
*The Serre-Swan Theorem and K-theory* *01/2021*

**Math-Phys-Cat Group**
*An invitation to model categories* *10/2020*

## Teaching experience

**Calculus II**
*Instructor of record, Mathematics Department, University of Virginia* *Fall 2026*

**Calculus II**
*Instructor of record, Mathematics Department, University of Virginia* *Fall 2025*

**A Survey of Calculus I**
*Instructor of record, Mathematics Department, University of Virginia* *Fall 2024*

**Calculus I**
*Instructor of record, Mathematics Department, University of Virginia* *Spring 2024*

**A Survey of Calculus II**
*Instructor of record, Mathematics Department, University of Virginia* *Fall 2023*

**Calculus II**
*Teaching assistant, Mathematics Department, University of Virginia* *Spring 2023*

**Calculus III**
*Graduate teaching assistant, Mathematics Department, University of Virginia* *Fall 2022*

**Differential Equations A**
*Graduate teaching assistant, Dep. of Mathematics, ICEx, UFMG* *10/2021-01/2022*

**Analytic Geometry and Linear Algebra**
*Graduate teaching assistant, Dep. of Mathematics, ICEx, UFMG* *05/2021-09/2021*

## Mentoring experience

**University of Virginia Math Experimental Lab**

*Co-mentor: Thomas Jaklitsch. Project: Compass and Straightedge Constructions.* *Spring 2026*

**University of Virginia Directed Reading Program**

*Mentee: Zubin Kaul. Project: Differential geometry of curves and surfaces.* *Fall 2024*

*Mentee: Mary Edith Plunkett. Project: Nonlinear dynamics and chaos.* *Spring 2024*

*Mentee: Brian Meiser. Project: Hamiltonian mechanics and Poisson brackets.* *Fall 2023*

## Conferences, schools, and workshops attended

2026 Fall Eastern Sectional Meeting, Washington, DC, USA *2026*

Union College Mathematics Conference, Schenectady, New York, USA *2026*

International Congress of Mathematicians, Philadelphia, Pennsylvania, USA *2026*

DV60: Discoveries and Vistas, Johns Hopkins University *2026*

Talbot Workshop 2026, Rangeley, Maine, USA *2026*

Atlantic TQFT Spring School 2026, Dalhousie University *2026*

BTW6: The Volume Conjecture, Frejus, France *2026*

Hgher Differential Geometry, Greifswald, Germany *2026*

GSTGC 2026, Rutgers University University *2026*

BUGCAT Conference, Binghamton University *2025*

Mathematical Congress of the Americas 2025, Miami, Florida, USA *2025*

Mid-Atlantic Collaborative Workshop in Algebraic Topology, Lexington, Virginia, USA *2025*

Atlantic TQFT Spring School 2025, University of New Brunswick *2025*

AMTRaK Conference, University of Pennsylvania *2025*

Mid-Atlantic Topology Conference, University of Virginia *2025*

New England Algebraic Topology and Mathematical Physics, Providence College *2024*

Encuentro Colombiano de Geometría y Topología, National University of Colombia *2024*

Atlantic TQFT Spring School 2024, Memorial University Newfoundland *2024*

Graduate Student Topology and Geometry Conference, Michigan State University *2024*

BUGCAT Conference, Binghamton University *2023*

Algebraic K-Theory and Redshift, Johannes Gutenberg University Mainz *2023*

Topological Moonshine, University of Illinois, Urbana-Champaign *2023*

Atlantic TQFT Spring School 2023, Wolfville, Nova Scotia, Canada *2023*

Homotopy Theory in Honor of Paul Goerss, Northwestern University *2023*

Field of Dreams Conference, Minneapolis, Minnesota, USA *2022*

I Mathematics Meeting of Minas Gerais, virtual participation due to COVID-19 *2021*

First Brazilian Meeting on Category Theory, virtual participation due to COVID-19 *2021*

Toposes online, virtual participation due to COVID-19 *2021*

## Scholarships

*Academic Excellence Program* (*PROEX*) *2020-2022*
M.Sc. program scholarship.

*Program of Undergraduate Research Scholarships* (*CNPq*) *2018-2019*
Undergraduate research scholarship.

## Academic service

Member of the directed reading program (DRP) committee *Fall 2026 - Present*

President of the University of Virginia AMS Graduate Student Chapter *Fall 2024 - Spring 2026*

## Languages

**Portuguese**: Native

**English**: Fluent

**Spanish**: Advanced

**French**: Intermediate